NOT MEASUREMENT SENSITIVE
MIL-P-85856/23
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS NICKEL-CHROME-ALUMINUM-IRON-MOLYBDENUM, TYPE I, CLASS 2, COMPOSITION DD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Mo	Cr	Ni	Fe	Al	Mn	Si	Others	
							Each	Total
3.0 8.0	11.0 16.0	60.0 min.	4.0 9.0	5.0 10.0	1.0 max.	1.0 max.	-	3.5 max.

Particle Size/Distribution

+100 Mesh (+150 µm)	5.0 max.
+120 Mesh (125 µm)	10.0 max.
+325 Mesh (+45 µm)	90.0 max.
-325 Mesh (-45 µm)	10.0 max.

Metallographic Inspection

Requires statement from powder manufacturers.

Sprayed Coating Properties

Bond Strength	20.7 MPa (3000 psi) min.
Hardness	Rockwell B, 75 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 15% max. Unreacted Particles - 15% max.

AMSC N/A FSC 3439

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Deposit Efficiency and Deposition Rate

60% min for N_2/H_2 @ 25.0 lb/hr
60% min for A/H_2 @ 12.0 lb/hr

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/23)

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